DATALabKU at ImageEval 2026 Shared Tasks: Metric-Aware Cultural Evaluation for CRAI-Bench

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Abstract

We present a reference-based evaluation framework for CRAI-Bench, Task 2 of ImageEval 2026. The method uses GPT-5.4 to extract cultural evidence from the reference image, caption, and generated image. This evidence is converted into metric-specific numerical features and mapped to CRAI scores using a small regressor trained on the released data. On the development set, the proposed approach improves composite Spearman correlation from 0.7505 to 0.8241, with the largest gains on CEA and CI. On the hidden test set, it achieves 0.8040 Spearman and 0.1415 MAE, ranking third overall and obtaining the lowest MAE among submitted systems. We also analyze performance across caption versions and show that evaluation becomes less reliable for some dimensions as captions become more generic.

1 Introduction

Text-to-image and vision-language models can generate and evaluate increasingly realistic visual content, including scenes that represent people, places, and cultural practices. However, visual alignment with the prompt does not guarantee cultural accuracy. Generated images may discard important cultural elements, replace them with incorrect alternatives, or combine elements from different cultures. Prior work has similarly shown variation in cultural understanding and representation across models depending on the region (Nayak et al., 2024, 2025; Tan et al., 2026).

These errors are also difficult to capture with conventional image-quality or image–text alignment metrics, especially when correctness depends on subtle cultural details. CRAI-Bench (Abdaljalil et al., 2026) addresses this problem by requiring fine-grained evaluation of generated images across several cultural dimensions using both reference and generated content.

We propose a reference-based evaluation framework based on structured cultural feature extraction. The framework identifies culturally relevant evidence from the reference image and caption, evaluates how this evidence is preserved in the generated image, and converts the resulting judgments into metric-specific numerical features. These features are then mapped to CRAI scores using a small regressor trained on the provided labeled data.

2 Related Work

Cultural understanding and image generation. Recent benchmarks have shown that multimodal models vary in their ability to represent culture-specific knowledge (Nayak et al., 2024; Tan et al., 2026). CulturalFrames further shows that existing image-generation metrics can disagree with human cultural judgments (Nayak et al., 2025), while CAIRE evaluates cultural relevance by grounding visual concepts in external knowledge (Yayavaram et al., 2026). In contrast, CRAI-Bench requires continuous evaluation across several cultural dimensions using both reference and generated images.

Automatic image evaluation. Image evaluation methods such as CLIPScore measure global image–text similarity (Hessel et al., 2021), while TIFA performs more fine-grained evaluation by generating and verifying questions from the prompt (Hu et al., 2023). These methods mainly assess prompt alignment and do not directly measure whether culturally important elements are preserved relative to a reference image. This motivates a reference-grounded evaluation that considers cultural details and their consistency across multiple scoring dimensions.

3 Task and Dataset

CRAI-Bench is Task 2 of ImageEval 2026 and focuses on predicting the cultural accuracy of text-to-image outputs (Abdaljalil et al., 2026). Each

Proceedings of The Fourth Arabic Natural Language Processing Conference (ArabicNLP 2026) - Volume 2 Shared Tasks, pages 759–765
October 28-29, 2026 ©2026 Association for Computational Linguistics

sample contains a real reference image of a Qatari cultural scene, an English caption used for image generation, and the corresponding generated image. Systems predict five CRAI dimensions, each assigned a weight in the final composite score. The five CRAI dimensions are Cultural Element Accuracy (CEA, 0.30), which measures whether the expected cultural elements are present and correctly depicted; Contextual Coherence (CC, 0.20), which evaluates whether these elements appear in an appropriate cultural manner; Cultural Specificity (CS, 0.20), which measures how closely the image represents Qatari culture; Cultural Integrity (CI, 0.20), which evaluates whether the representation remains free from cultural distortion; and Hallucination Penalty (HP, −0.10), which captures fabricated elements that are not supported by the caption. Each dimension is assigned a score between 0 and 1. The composite score is calculated as:

$$\begin{aligned}\mathrm{CRAI} = 0.30\,\mathrm{CEA} + 0.20\,\mathrm{CC} + 0.20\,\mathrm{CS} \\ + 0.20\,\mathrm{CI} - 0.10\,\mathrm{HP}.\end{aligned} \quad (1)$$

The dataset contains 200 instances derived from 40 reference images covering people and traditional clothing, architecture and built environments, and cultural objects (Abdaljalil et al., 2026). Each image is paired with five captions ranging from very Qatari-specific to more generic and concise descriptions. The data are split by reference image into 120 training, 40 development, and 40 test instances. Human scores are available for training and development, while the test labels remain hidden. Systems are evaluated using Spearman correlation as the primary measure and mean absolute error (MAE) as the secondary measure, both for the composite score and each CRAI dimension.

3.1 System Overview

Our system uses a frozen multimodal GPT-5.4 model to perform reference-grounded cultural evaluation. Rather than asking the model to directly predict a CRAI score, we first extract structured evidence describing the cultural elements in the reference image and how they are preserved or changed in the generated image. This evidence is then converted into numerical features and mapped to the final metric score. Figure 2 summarizes the overall pipeline.

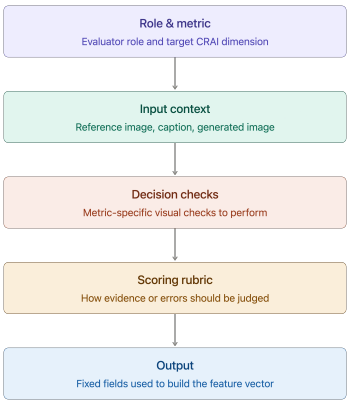


Figure 1: Prompt structure used for CRAI evaluation.

3.1.1 Cultural Anchor Extraction

The model first uses the reference image and caption to identify important cultural details, which we call *cultural anchors*. These may include objects, clothing details, or distinctive visual attributes. GPT-5.4 is prompted as a *strict multimodal CRAI-Bench evaluator* and is instructed to focus on the target metric. As illustrated in Figure 1, the prompt defines the role of each input, provides metric-specific checks and an evaluation rubric, which we define by manually inspecting a few samples in the train set, and requires a fixed structured output.

The model then evaluates each cultural anchor in the generated image. It checks whether the anchor is present, preserves its identity, is represented correctly, or has been replaced or contradicted. The resulting structured evidence is then used in the following stage.

3.1.2 Reference-Based Feature Extraction

Direct scoring asks the model to summarize several visual comparisons in one number. This can hide small cultural errors. An image may appear reasonable while replacing a culture-specific element with a generic or culturally incorrect alternative. Inspired by LLMScore (Lu et al., 2023) and CAIRE (Yayavaram et al., 2026), we use GPT-5.4 to extract structured evidence rather than relying on a direct final judgment.

First, the model uses the reference image and caption to identify the important cultural details, which we call *cultural anchors*. These may include objects, clothing details, or unique visual attributes. The model then checks each anchor in the generated image. It determines whether the anchor is present, whether it remains the same element,

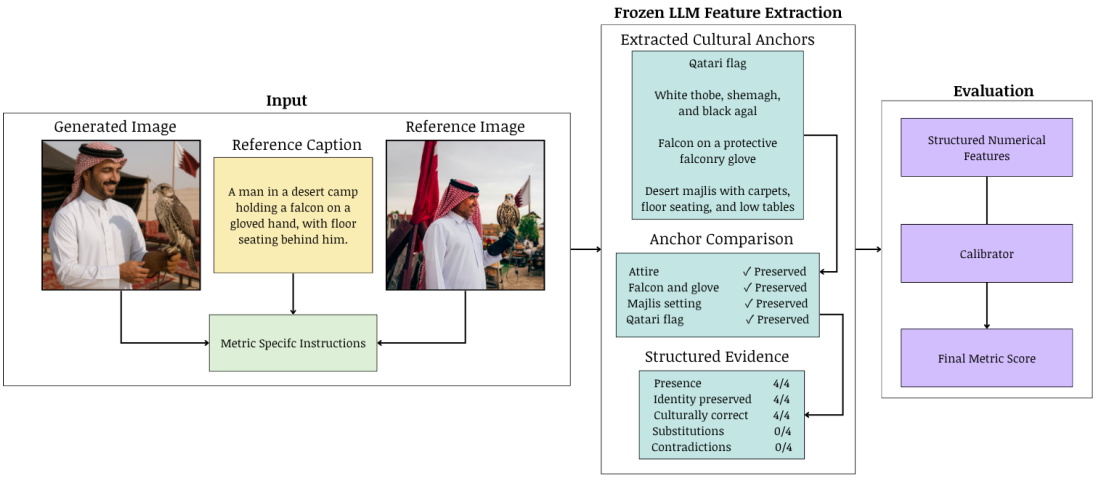


Figure 2: Overview of the proposed reference-based evaluation framework. Cultural anchors are identified from the reference image and caption, compared with the generated image to assess preservation, substitution, and cultural correctness, and then converted into numerical features for final score prediction.

whether it is shown correctly, and whether it has been replaced or contradicted. These checks separate different errors, such as discarding an element, replacing it with a generic alternative, or showing it in a culturally incorrect form.

3.1.3 Feature Regression and Score Calibration

The structured evidence produced by the evaluator is converted into a fixed-length numerical feature vector. These features capture aspects such as the overall metric assessment, the evaluator's confidence, and the degree to which relevant cultural evidence is preserved or violated. The resulting values are concatenated to form a fixed-length numerical feature vector. For a given metric, the feature vector is represented as

$$\mathbf{x}_m = [f_{m,1}, f_{m,2}, \ldots, f_{m,d_m}], \qquad f_{m,i} \in [0,1] \tag{2}$$

where m denotes the cultural metric and d_m is the number of features used for that metric. For example, CEA includes the overall CEA judgment and confidence, together with features describing how well the identified cultural elements are preserved, such as their preservation score and the presence of generic substitutions. The resulting feature vector is mapped to a single metric score using ridge regression:

$$\hat{y}_m = g_m(\mathbf{x}_m; \theta_m) \tag{3}$$

where g_m denotes the learned score-mapping function. Ridge regression is evaluated with $\alpha \in \{0.1, 1, 10, 100\}$ using nested group-based cross-validation, with five outer folds and four inner folds. Samples are grouped by reference image so that all caption variants associated with the same scene remain in the same fold. This prevents closely related samples from appearing in both the training and validation partitions.

After configuration selection, the selected regressor is fitted using the training data and fixed for inference. For an unseen sample, the structured evidence is converted into its corresponding feature vector $\mathbf{x}_m$, which is then passed through the fitted regressor to obtain the final metric score $\hat{y}_m$.

4 Experimental Setup

The feature-based scoring configuration for each CRAI metric is selected using the 120 labeled training instances. Performance is measured primarily by mean Spearman correlation and MAE as a secondary criterion.

We compare the proposed reference-based feature extraction approach against a direct-scoring baseline, where GPT-5.4 predicts the target CRAI score directly from the reference image, generated image, and caption. The development set is used to report the performance of each approach for every CRAI metric. All experiments use only the released task data.

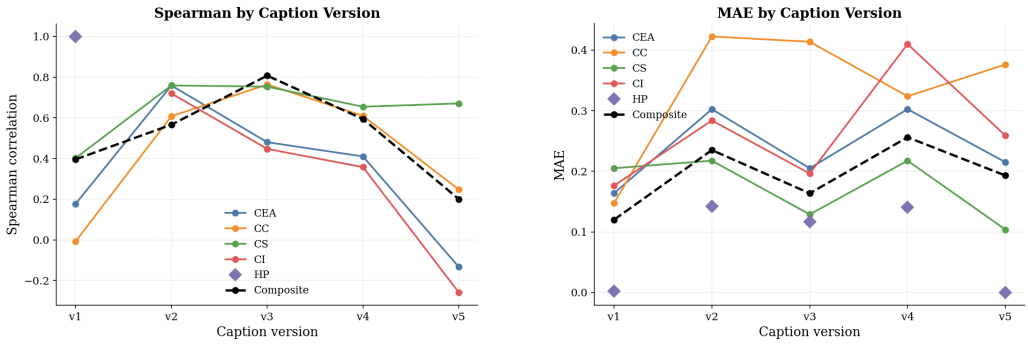
5 Results

5.1 Development Results

Table 1 summarizes the results for the proposed approach and direct baseline. It achieves the strongest

Table 1: Development-set performance of the proposed approach and direct GPT-5.4 scoring.

Metric	Spearman		MAE	
	Our Approach	Direct GPT-5.4	Our Approach	Direct GPT-5.4
CEA	**0.7732**	0.6950	**0.2303**	0.2378
CC	0.7306	**0.7352**	**0.2423**	0.3368
CI	**0.8276**	0.7043	**0.1583**	0.2650
CS	0.7738	**0.8184**	0.1969	**0.1745**
HP	0.1188	**0.1388**	0.1563	**0.1347**
Composite	**0.8241**	0.7505	0.1943	**0.1893**



(a) (b)

Figure 3: Performance across caption versions on the development set: (a) Spearman correlation and (b) MAE for the five CRAI dimensions and the composite score.

composite Spearman correlation, reaching 0.8241 compared with 0.7505 for direct scoring. At the individual-metric level, it performs best on CEA and CI for both Spearman and MAE. For CC, the two approaches obtain similar Spearman correlation, while the proposed approach reduces MAE from 0.3368 to 0.2423. On the other hand, direct scoring performs better on CS and HP, and also obtains a slightly lower composite MAE of 0.1893 compared with 0.1943. HP remains the weakest dimension for both approaches, with Spearman correlation below 0.14.

5.2 Development Analysis

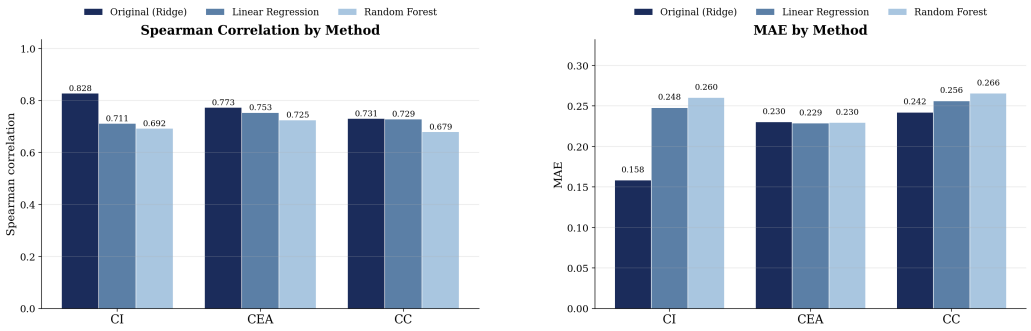
Figure 3 shows that performance changes with caption specificity. The detailed (v1) captions contain many explicit cultural cues. For example, one (v1) caption describes "a man standing in a desert majlis camp, holding a falcon ... wearing white thobe with shemagh ... [with] carpets, low tables, and a visible Qatari flag," giving the evaluator several concrete details to compare with the reference image. These detailed captions generally produce low absolute errors.

The best performance is observed for the intermediate (v2)–(v3) captions, with the composite Spearman correlation peaking at (v3). As the captions become shorter and more generic toward (v5), much of the cultural detail is removed. For the same example, the (v5) caption is reduced to "A person holding a bird outdoors." With fewer explicit cues, it becomes harder to determine which cultural details should influence the score, and the correlations for CEA and CI decrease sharply. CS remains relatively stable, which indicates that cultural specificity can still be maintained even when the caption provides little cultural detail.

Figure 4 compares the original score mapping with Linear Regression and Random Forest for CI, CEA, and CC. The original mapping generally achieves higher Spearman correlation and lower MAE, with the largest difference observed for CI.

Table 2: Official ImageEval 2026 Task 2 hidden-test leaderboard.

Rank	System	Spearman ↑	MAE ↓
1	Submission 1	**0.8258**	0.1469
2	Submission 2	0.8078	0.1512
3	**DATALabKU (Ours)**	0.8040	**0.1415**
4	Submission 4	0.7806	0.1452
5	Submission 5	0.6734	0.2032
–	GPT-4o Judge Baseline	0.5186	0.2357



(a) (b)

Figure 4: Comparison of the original score mapping with Linear Regression and Random Forest for CI, CEA, and CC: (a) Spearman correlation and (b) MAE on the development set.

For CEA, the three methods obtain almost identical MAE, although the original mapping provides the highest correlation. For CC, Linear Regression achieves nearly the same correlation as the original mapping, but with a higher MAE.

5.3 Hidden-Test Results

As for the hidden-test results, they are reported in Table 2. DATALabKU ranked third with a composite Spearman correlation of 0.8040, only 0.0038 below the second-ranked submission. Although ranked third according to the primary Spearman metric, our system achieves the lowest MAE among all submitted systems, reaching 0.1415. It also substantially outperforms the GPT-4o judge baseline on both metrics, showing better ranking agreement with the human scores while maintaining low absolute prediction error.

6 Conclusion

We presented a reference-based feature extraction framework for evaluating cultural representation in CRAI-Bench. Instead of relying only on direct multimodal scoring, the proposed approach extracts structured cultural evidence from the reference and generated images and converts it into metric-specific numerical features. On the development set, the approach improves the composite Spearman correlation from 0.7505 with direct GPT-5.4 scoring to 0.8241, with particularly strong gains for CEA and CI.

The results also show that performance depends on caption specificity, with the strongest ranking performance generally observed for intermediate caption versions and weaker results for some metrics when captions become highly generic. HP remains the most difficult dimension for both approaches. Future work will further investigate hallucinations and ways to better detect them, and make the scoring framework more robust when the caption provides little explicit cultural information.

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A Metric-Specific Prompts

This section summarizes the metric-specific instructions and outputs used by the evaluators.

A.1 Cultural Element Accuracy (CEA)

Prompt. The evaluator identifies the main culture-bearing elements from the reference image and caption and checks whether they remain present, recognizable, and culturally correct in the generated image.

Numerical rubric. Each cultural element is evaluated using the following scale:

- 1.00: clearly preserved and culturally correct;
- 0.75: recognizable with minor nonessential differences;
- 0.50: partly recognizable, incomplete, or culturally generic;
- 0.25: weak resemblance with major cultural errors or an incorrect subtype;
- 0.00: absent, replaced by a generic or noncultural substitute, or culturally incorrect.

Output fields. `raw_cea`, `confidence`, `caption_scope`, and cultural features describing presence, identity match, cultural correctness, and substitution with generic alternatives.

A.2 Contextual Coherence (CC)

Prompt. The evaluator judges whether the visible cultural elements form a coherent scene, considering their placement, relationships, actions, and functions.

Numerical rubric. The evaluator uses five score levels:

- 1.00: fully coherent cultural setting and relationships;
- 0.75: mostly coherent, with only minor contextual problems;
- 0.50: partly coherent, with noticeable contextual problems;
- 0.25: major contextual conflicts;
- 0.00: culturally or functionally incoherent.

Output fields. The output contains probabilities for setting information and Qatari-context status.

A.3 Cultural Specificity (CS)

Prompt. The evaluator checks whether reference-grounded cultural statements remain supported by the generated image and whether the visible evidence is aligned with the intended Qatari context.

Numerical rubric. Each statement is assigned one of three values:

- 1.00 (`true`): the statement is clearly satisfied;
- 0.50 (`partial`): the statement is only partly satisfied;
- 0.00 (`false`): the statement is not satisfied.

Output fields. `cultural_cue_presence`, `cultural_identity_preservation`, `qatar_specificity`, `generic_substitution`, `statement_satisfaction`.

A.4 Cultural Integrity (CI)

Prompt. The evaluator judges whether the cultural identity and meaning of the reference remain recognizable, and free from material distortion.

Numerical rubric. The score is interpreted using the following levels:

- 1.00: cultural identity and meaning are preserved without meaningful distortion;
- 0.75: the core cultural meaning is preserved with only minor inaccuracies;
- 0.50: the broad identity remains, but important cultural meaning is weakened;
- 0.25: major substitutions or distortions weaken the intended identity;
- 0.00: the intended cultural identity is lost or seriously distorted.

Output fields. `core_identity_retention`, `practice_or_form_integrity`, `respect_and_plausibility`, `generic_substitution_severity`, `material_distortion_severity`, `integrity_failure`, `raw_ci`, and `confidence`.

A.5 Hallucination Penalty (HP)

Prompt. The evaluator searches for distinct generated elements that are unsupported by both the current caption and the reference image, such as fabricated additions.

Numerical rubric. Hallucination severity is interpreted as:

- 0.00: no eligible generated-only cultural fabrication;
- 0.10–0.25: subtle or small extra entity, duplication, or misleading marker;
- 0.30–0.50: noticeable fabricated element that materially changes the scene;
- 0.55–0.70: major culturally misleading fabricated element;
- 0.75–1.00: fabricated content dominates or severely corrupts the cultural meaning.

Output fields. `verdict`, `candidate_type`, `caption_support`, `reference_support`, `generated_only`, `eligible_fabrication`, `severity`, `confidence`, `visible_evidence`, and `brief_reason`.

B Limitations

The proposed framework relies on GPT-5.4, so its behavior may vary with future model or API updates. In addition, Hallucination Penalty remains difficult to predict and requires further investigation. Due to computational constraints, the regressor analysis is limited to CEA, CC, and CI, and does not include CS or HP.

C Ethics Statement

This work uses only the data released as part of ImageEval 2026 and does not collect new personal or sensitive information. The framework is intended to evaluate cultural representation and not make judgments about certain cultures.

D AI Assistance Disclosure

Generative AI tools were used to assist with language editing and improve the clarity of the presentation. All experimental design, implementation, analyses, figures, and conclusions were developed, verified, and approved by the authors.